

Congressional bipartisanship scores by member and issue area, 1983 – 2024

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ABSTRACT

Although bipartisanship is central to the study of legislative behavior and often necessary for policymaking success, scholars lack a centralized, publicly accessible, and consistently updated dataset measuring legislators' propensity to offer and attract cross-party support over time and across issue areas. As a result, researchers often construct different measures of bipartisanship, making it harder for scholars to build on existing work and engage in cumulative conversations about whether bipartisanship is declining, who engages in it, and which issues facilitate cross-party collaboration. We introduce a comprehensive dataset of congressional sponsorship and cosponsorship activity for the U.S. House and Senate from 1983 to 2024 and use these data to construct Congressional Bipartisanship Scores (CBS). Built from more than 2.4 million cosponsorship decisions on 147,669 bills, CBS provides two member-term measures: attracting original out-party cosponsors to one's bills and offering original cosponsorship to out-party-sponsored bills. The dataset includes aggregate and issue-specific scores across 34 policy areas for 2,056 unique legislators and 11,549 legislator-term observations. To facilitate broad reuse, we make both the underlying sponsorship and cosponsorship data and the resulting scores publicly available through Harvard Dataverse and the R package `biparty`.

Background & Summary

Bipartisanship plays a central role in congressional lawmaking. In an institutional environment structured by bicameralism, separation of powers, and multiple veto points, the capacity of legislators to work across party lines is not only a normative ideal [1, 2], but often a necessary pathway for advancing and passing legislation [3, 4]. Legislators, in turn, have strong governing and electoral incentives to cooperate across party lines. Bipartisan coalitions can reduce resistance from pivotal actors, facilitate committee progress, and increase the likelihood that legislation reaches the floor [5, 6]. As a result, legislators who attract bipartisan support advance a greater share of their sponsored bills into law while broader coalitions can produce more durable legislation and expand the representational reach of policy [7–9]. Voters, particularly in general elections, often respond favorably to legislators who cultivate bipartisan reputations [10–13]. And candidates appear to recognize these incentives, frequently campaigning on promises of bipartisan lawmaking and subsequently engaging in more bipartisanship once elected [13–16].

Yet in a polarized Congress, bipartisanship has become increasingly risky [17–19]. Although cross-party cooperation may help legislators govern and appeal to general-election constituencies, it can also impose significant partisan costs. Intense competition for majority control gives party leaders stronger incentives to deny the opposing party legislative accomplishments and preserve clear partisan distinctions [20], making cross-party collaboration politically costly. Likewise, primary voters can punish legislators electorally for compromising with the opposing party [21]. This changing political calculus heightens the need for comprehensive data and careful measurement of bipartisanship in Congress.

Existing work measures congressional bipartisanship in various ways, each capturing a substantively different form of cross-party behavior. Roll-call voting is a common measure because it identifies whether legislators ultimately cross party lines when voting on legislation that reaches the floor [22–25]. Its principal limitation, however, is selection: recorded votes occur only after committees and party leaders have already shaped which proposals reach the floor [22]. As a result, observed roll-call bipartisanship captures legislators' willingness to cross party lines on only a limited, nonrandom set of legislation. This concern is particularly relevant in the contemporary Congress because parties have incentives to structure the floor agenda around partisan legislation that differentiates them, meaning roll-call voting may understate bipartisan cooperation that occurs earlier in the legislative process [22]. Senate amendment cosponsorship provides another potential window into cross-party collaboration, but it faces a related selection problem: amendments are not offered to every bill. Moreover, because formal

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amendment cosponsorship is a Senate-specific behavior, it cannot provide a comparable measure across both chambers.¹

Bill cosponsorship avoids many of these limitations and, as a result, has become the most widely used measure of bipartisanship in legislative studies [7, 14, 22, 30–34]. It offers several measurement advantages. First, cosponsorship captures bipartisan behavior before committees and party leaders have narrowed the set of proposals that reach the floor. As a result, cosponsorship-based measures can identify whether legislators choose to work with the out party across the full universe of introduced bills. Second, cosponsorship is directional: legislators can offer support to bills sponsored by members of the opposing party, while sponsors can attract support from out-party legislators. These behaviors capture conceptually distinct dimensions of bipartisanship [7]. Third, cosponsorship data are temporal, allowing resulting measures to consider when legislators enter a coalition and distinguish those who help initiate bipartisan coalitions by coauthoring legislation with out-party members from those who join after a bipartisan coalition has already formed [19].

Despite the measurement advantages of using sponsorship and cosponsorship to study bipartisan collaboration, important data limitations make it difficult to measure this behavior consistently across chambers, time, and policy areas. No existing publicly available dataset provides broad longitudinal coverage of legislators' bipartisan cosponsorship activity across both the attract and offer dimensions and across substantive policy areas. Existing resources address portions of this need: some data provide sponsorship and cosponsorship records across earlier Congresses [35], while others distinguish between legislators' ability to attract bipartisan support and their willingness to offer it [7]. The Lugar Center's Bipartisan Index [36] provides the most prominent publicly available member-level measure of congressional bipartisanship, but it has several important limitations. First, the index counts cross-party cosponsorship regardless of when a legislator joins a bill and does not distinguish between members who participate in a bill's initial bipartisan coalition and those who join later. Second, it combines bipartisan sponsorship and cosponsorship into a single composite score rather than measuring legislators' propensity to attract and offer cross-party support separately. Third, the index combines five standardized metrics—including counts, percentages, and the intensity of bipartisan support—and evaluates members relative to historical benchmarks, making the resulting scores less directly interpretable as quantities of observed cross-party behavior. Fourth, its temporal coverage differs substantially across chambers, beginning with the 103rd Congress for the Senate and the 113th Congress for the House. Fifth, the Lugar Center does not provide bipartisanship scores by policy area. Finally, its coverage excludes some legislators, including top chamber leaders, while minimum-service requirements can also exclude members who switch chambers during a Congress. Collectively, existing limitations across data sources make it difficult for researchers to distinguish different forms and stages of bipartisan coalition-building, compare those behaviors consistently across the House and Senate over long periods, or examine how bipartisanship varies across policy domains.

In this article, we make two primary data contributions. First, we introduce a comprehensive, publicly available dataset of congressional sponsorship and cosponsorship activity on 212,220 bills with more than 2.4 million cosponsorship decisions across both chambers for 21 congressional terms from 1983 to 2024. These underlying data provide researchers with a standardized, cleaned, and quality-controlled record of who sponsors and cosponsors legislation, when those decisions occur, and the policy areas they concern. Scholars can use the data to study a wide range of questions about legislative behavior, networks, and coalition formation, including but not limited to bipartisanship.

Second, we use these data to construct the Congressional Bipartisanship Scores (CBS), a set of member-term measures designed to address limitations in existing measures of bipartisanship. First, CBS measures bipartisanship using original cosponsorship—out-party support offered on the date a bill is introduced. Original cosponsorship captures participation in a bill's initial coalition, including drafting, negotiation, and coauthoring that can occur before introduction [33]. Second, CBS provides separate measures of the out-party support legislators attract to the bills they sponsor and the support they offer to bills sponsored by out-party members. Third, CBS provides comparable measures for the House and Senate across every Congress from the 98th through the 118th. Fourth, we provide both aggregate and issue-specific scores across 34 Congressional Research Service (CRS) policy areas. Fifth, the scores are directly interpretable as proportions of observed cross-party support, with both raw and precision-weighted versions available. Finally, CBS retains chamber leaders and includes multiple member-level identifier crosswalks to facilitate merging with other congressional datasets. In total, the scores cover 2,056 unique legislators and 11,549 legislator-term observations.

¹Senators may formally cosponsor amendments, making amendment cosponsorship a potential measure of bipartisan collaboration. Existing research has used amendment sponsorship and cosponsorship to study legislative preferences and patterns of legislative collaboration [26–28], although it has not generally been used to construct broad measures of congressional bipartisanship. House members cannot formally cosponsor amendments in the same manner [29]. Amendment cosponsorship may therefore be useful for studying cross-party collaboration within the Senate, but it cannot provide a comparable measure across both chambers or the full universe of introduced legislation.

We validate the underlying data by hand-coding 5,000 randomly sampled cosponsor-bill pairs and assess external validity by comparing our scores with the Lugar Center’s Bipartisan Index [36]. We make the data, replication code, and documentation publicly available through Harvard Dataverse [37] and the R package *biparty* [38]. Together, these resources provide standardized, open, and replicable measures of aggregate and issue-specific bipartisanship while allowing researchers to distinguish when legislators offer and attract cross-party support. The article details the data collection, processing, score construction, technical validation, data structure, and visualization tools included in *biparty*.

Methods

We begin by constructing a comprehensive dataset of congressional sponsorship and cosponsorship activity based on the universe of 212,220 House and Senate bills introduced from 1983 through 2024. The dataset combines bill, sponsor, cosponsor, and policy-area information from *congress.gov* and records more than 2.4 million cosponsorship decisions, and we make these underlying data publicly available. We then use these data to construct CBS, which include both aggregate and issue-specific measures of the bipartisan support legislators attract and offer. Figure 1 summarizes the data-construction process and the exclusions, recoding decisions, and sample restrictions used in constructing the scores.

Collecting Bill, Sponsor, Cosponsor, and Policy Issue Area Data

First, we scraped bill, sponsor, cosponsor, and policy issue area data from the *congress.gov* Application Programming Interface (API), the official public congressional record maintained by the Library of Congress, for the 98th through the 118th Congresses (1983–2024). We collected these data between December 2025 and February 2026 using the following *congress.gov* API endpoints: `/bill/congress/billType/billNumber/text`, `/member/bioguideId/sponsored-legislation`, `/member/bioguideId/cosponsored-legislation`, and `/bill/congress/billType/billNumber/policy area`. We collected information on the full universe of 212,220 introduced House and Senate bills—that is, measures with the H.R. and S. designations—regardless of whether they attracted any cosponsors. We excluded resolutions, joint resolutions, and concurrent resolutions because many serve symbolic, procedural, or internal chamber purposes. Focusing on bills allows us to observe sponsorship and cosponsorship on the primary legislative vehicle for proposing changes to law. We organize the collected information into four preliminary data frames: bills, sponsors, cosponsors, and policy issue areas.

The first data frame, *bills*, is organized at the bill-term level and contains all 212,220 introduced H.R. and S. bills. For each bill, we retain the bill number, title, URL, introduction date, and total number of cosponsors. The second data frame, *sponsors*, is organized at the bill-sponsor-term level and contains information on the legislator who introduced each bill, including the sponsor’s unique identifier, demographic characteristics, party identification, and introduction date. Because each bill has a single sponsor, the *sponsors* data frame also contains 212,220 observations.

The third data frame, *cosponsors*, is organized at the cosponsor-bill-term level and contains at least one observation for every unique bill. Bills that attract no cosponsors are retained in the dataset with missing cosponsor-related information. For bills that attract one or more cosponsors, each observation represents a unique cosponsorship decision, with each row containing information on a single cosponsor. As a result, a bill appears multiple times in the dataset if it has multiple cosponsors. The data frame includes each cosponsor’s unique identifier, demographic characteristics, party identification, and the date the legislator joined the bill. In total, the data frame contains 2,550,640 observations. Across the 147,669 bills that attracted at least one cosponsor, we observe 2,486,478 cosponsorship decisions.

The fourth data frame, *policy issue areas*, is organized at the bill-term level. The Congressional Research Service (CRS) assigns each bill to one of 34 policy areas, and *congress.gov* links these classifications to individual bills. When scraping the API, 374 bills lacked an assigned policy area. We manually searched each of these bills on *congress.gov* to determine whether a policy area was listed on the website itself. For 41 bills, we identified a policy area and updated the data accordingly. For the remaining 333 bills, no policy area was available on *congress.gov*, so we code their policy areas as NA and retain the bills in the data. Over our time series, the number of available policy areas decreased from 34 to 32, as CRS no longer assigns bills to the categories “Commemorations” (after the 110th Congress) or “Private Legislation” (after the 113th Congress). We retain bills assigned to these categories during the periods in which they were used.²

²Because some scholars may prefer to use the Comparative Agendas Project (CAP) policy area coding [39] rather than the Congressional Research Service (CRS) policy area labels, we provide a crosswalk between the two coding schemes in a separate raw data file included with the replication materials. However, CAP bill-level policy coding extends only through 2016. Given this substantial gap in temporal coverage, our released data use CRS policy area labels throughout.

We then merge the four preliminary data frames to create the underlying sponsorship and cosponsorship dataset. We first merge *sponsors* and *policy issue areas* into *bills* at the bill-term level, preserving all 212,220 introduced H.R. and S. bills. We then merge the cosponsorship records into this bill-level data using a one-to-many join. Bills with cosponsors consequently appear once for each observed cosponsorship relationship, while bills without cosponsors remain as a single bill-level observation with missing cosponsor information. The resulting merged data frame preserves the complete universe of introduced H.R. and S. bills while organizing observed cosponsorship relationships in long format.

Sponsorship and Cosponsorship Data

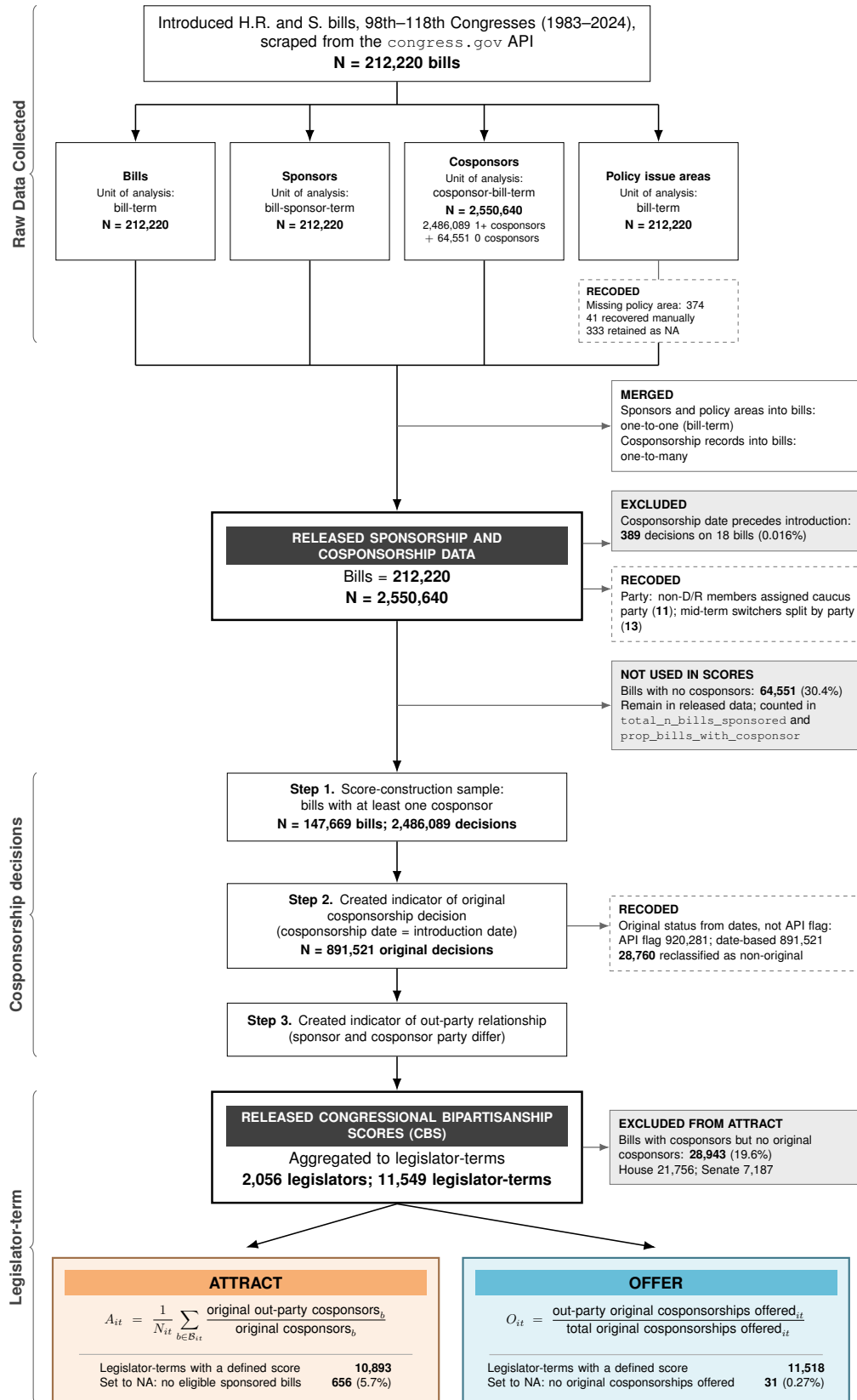
We make the resulting merged data frame publicly available as a standalone contribution of the project. The data provide a standardized record of congressional bill sponsorship and cosponsorship across the House and Senate from the 98th through the 118th Congresses (1983–2024). The dataset contains 2,550,640 observations in long format, preserving all 212,220 introduced H.R. and S. bills and their valid cosponsorship relationships. For each bill, the data identify its sponsor, any cosponsors, the timing of those legislative decisions, and the policy area of the legislation. Researchers can, in turn, use these data independently of CBS to study sponsorship and cosponsorship behavior, legislative networks, coalition formation, the timing of legislative support, among many other dimensions of congressional behavior. Importantly, because the underlying data retain all introduced H.R. and S. bills, including those that do not enter our score construction because they attract no cosponsors, researchers can impose alternative sample restrictions or construct measures of bipartisanship based on different theoretical and measurement choices.

We apply several additional cleaning and preprocessing steps to the underlying sponsorship and cosponsorship records, summarized in Figure 1. First, we identify 18 bills containing 389 cosponsorship decisions for which the recorded cosponsorship date precedes the bill's sponsorship date, representing 0.0156% of all cosponsorship decisions. We manually examined each of these bills and confirmed that the sponsorship date reported by `congress.gov` matches the introduction date embedded in the bill text and that, in every case, both dates occur after the recorded cosponsorship dates.³ Because a legislator cannot cosponsor a bill before it has been introduced, we exclude the 389 affected cosponsorship decisions from the released data. We also recode party identification to reflect legislators' functional partisan alignment within Congress. Legislators who are not formally identified as Democrats or Republicans are assigned to the major party with which they caucus or are functionally aligned; in total, this affects 11 legislators.⁴ Finally, we retain 13 legislators who switched parties during a congressional term and create separate observations corresponding to each party affiliation.

³The following 18 bills had at least one cosponsorship date that preceded the sponsorship date, all of which were manually verified: H.R. 4813 (98th), H.R. 5465 (98th), H.R. 4368 (98th), H.R. 3350 (98th), H.R. 6205 (102nd), H.R. 6 (106th), H.R. 4674 (109th), H.R. 4678 (109th), H.R. 4675 (109th), H.R. 4673 (109th), H.R. 4665 (109th), H.R. 4662 (109th), H.R. 4660 (109th), H.R. 4667 (109th), H.R. 4677 (109th), H.R. 4666 (109th), H.R. 4668 (109th), and H.R. 4655 (109th).

⁴We recode legislators who were not formally identified as Democrats or Republicans according to the major party with which they caucused or were functionally aligned in Congress. Specifically, we code Victor Frazer (I), Bernie Sanders (I), Dean Barkley (I), James Jeffords (I), Angus King (I), Joseph Lieberman (ID), Joe Manchin (I), and Kyrsten Sinema (I) as Democrats, and Justin Amash (I/L), Virgil Goode (I), and Baltasar Corrada (I) as Republicans. For legislators whose recorded affiliation changed during our period, we apply the recode only to observations in which they are listed by `congress.gov` as Independent (I), Independent Democrat (ID), or Libertarian (L).

Figure 1. Data Construction Pipeline for the Congressional Bipartisanship Scores



Note: The center column reports the observations retained at each stage of data construction; boxes on the right report observations excluded (shaded), recorded (dashed), or merged (solid) at that stage.

The released data contain bill-level information for every introduced H.R. and S. bill and cosponsor-level information for each observed cosponsorship relationship, including bill identifiers and titles, sponsor and cosponsor identifiers and characteristics, party affiliation, introduction and cosponsorship dates, and CRS policy area. Table 1 summarizes each variable in the merged dataset. For our purposes, we use the cleaned merged data frame described above to construct CBS.

Table 1. Variables from Sponsorship and Cosponsorship Dataset

Variable	Description	Example
<code>congress</code>	Congressional term in which the bill was introduced	117
<code>bill_type</code>	Type of legislation	S
<code>bill_number</code>	Bill number within the congressional term and chamber	5357
<code>title</code>	Official title of the bill	Justice for the Living Victims of Lockerbie Act
<code>bill_url</code>	<code>congress.gov</code> API URL for the bill	.../117/s/5357
<code>n_cosponsors</code>	Total number of legislators recorded as cosponsors of the bill	5
<code>has_cosponsor</code>	Indicator equal to <code>TRUE</code> if the bill attracted at least one cosponsor	<code>TRUE</code>
<code>sponsor_bioguide_id</code>	Unique Biographical Directory identifier for the bill sponsor	C001035
<code>sponsor_first_name</code>	Sponsor's first name	Susan
<code>sponsor_last_name</code>	Sponsor's last name	Collins
<code>sponsor_full_name</code>	Sponsor's full name and congressional party-state designation	Sen. Collins, Susan M. [R-ME]
<code>sponsor_party</code>	Sponsor's party identification	R
<code>sponsor_state</code>	State represented by the sponsor	ME
<code>sponsorship_date</code>	Date on which the bill was introduced	2022-12-22
<code>policy_area</code>	Congressional Research Service policy area assigned to the bill	Law
<code>policy_area_number</code>	Numeric identifier assigned to the bill's policy area in the dataset	14
<code>bill_id</code>	Unique bill identifier combining Congress, bill type, and bill number	117-s-5357
<code>cosponsor_bioguide_id</code>	Unique Biographical Directory identifier for the cosponsor	F000062
<code>cosponsor_first_name</code>	Cosponsor's first name	Dianne
<code>cosponsor_last_name</code>	Cosponsor's last name	Feinstein
<code>cosponsor_full_name</code>	Cosponsor's full name and congressional party-state designation	Sen. Feinstein, Dianne [D-CA]
<code>cosponsor_party</code>	Cosponsor's party identification	D
<code>cosponsor_state</code>	State represented by the cosponsor	CA
<code>cosponsorship_date</code>	Date on which the legislator joined the bill as a cosponsor	2022-12-22
<code>cosponsor_url</code>	<code>congress.gov</code> API URL for the cosponsor	.../member/F000062

Calculating Aggregate Bipartisanship Measures

From the merged dataset, we calculate aggregate bipartisanship measures at the member-term level in four steps. First, we restrict the score-construction sample to the 147,669 bills that attracted at least one cosponsor, excluding 64,551 bills (30.4%) for which no cosponsors were attracted. These excluded bills remain available in the underlying data, but a cosponsorship-based measure of bipartisanship cannot be calculated for bills with no cosponsors. The retained bills contain 2,486,089 cosponsorship decisions.

Second, we identify original cosponsorship directly from the recorded dates rather than the `congress.gov` original-cosponsor indicator. Specifically, we code a cosponsorship as original when the cosponsorship date matches the bill's introduction date. This approach reclassifies 28,760 cosponsorship decisions that `congress.gov` identifies as original even though the recorded cosponsorship dates occur after introduction.

Third, we create an indicator of an out-party relationship, coded 1 when a bill sponsor and cosponsor do not share the same party identification and 0 otherwise. This indicator allows us to distinguish cross-party from same-party cosponsorship ties.

Fourth, we construct two aggregate member-term measures of bipartisanship: *attracting* out-party cosponsorship to a legislator's bills and *offering* cosponsorship to bills sponsored by out-party members. The first measure, *attract*, captures the extent to which a legislator's sponsored bills attract cross-party support at introduction. For each bill that attracted at least one original cosponsor, we divide the number of original out-party cosponsors by the total number of original cosponsors. Bills that attracted no original cosponsors, including bills that subsequently attracted one or more later cosponsors, are excluded from the calculation of this measure. We then average this bill-level proportion across all qualifying bills sponsored by a legislator in a given term. Formally,

$$A_{it} = \frac{1}{N_{it}} \sum_{b \in \mathcal{B}_{it}} \left(\frac{\text{number of original out-party cosponsors on bill}}{\text{total number of original cosponsors on bill}} \right) \quad (1)$$

where:

- A_{it} = proportion of out-party original cosponsors attracted by legislator i in term t
- $\mathcal{B}_{it}$ = set of bills sponsored by legislator i in term t that attracted at least one original cosponsor
- N_{it} = number of bills in $\mathcal{B}_{it}$
- b = bill

The second measure, offer, captures the extent to which a legislator joins an out-party bill as an original cosponsor. For each legislator-term, we divide the total number of original out-party cosponsorships by the total number of original cosponsorships for bills sponsored by legislators of either party. Formally,

$$O_{it} = \frac{\text{out-party original cosponsorships offered}_{it}}{\text{total original cosponsorships offered}_{it}} \quad (2)$$

where:

- O_{it} = proportion of original cosponsorships offered by legislator i in term t that are offered to out-party-sponsored bills
- Out-Party Original Cosponsorships Offered_{it} = number of original cosponsorships legislator i offers to out-party-sponsored bills in term t
- Total Original Cosponsorships Offered_{it} = total number of original cosponsorships legislator i offers in term t

Together, these measures capture two distinct dimensions of aggregate bipartisanship at the member-term level: the extent to which legislators *attract* cross-party support for their sponsored legislation and the extent to which they *offer* cross-party support for bills sponsored by others. Legislators who did not introduce a bill in a term, or whose sponsored bills did not attract at least one original cosponsor, receive an NA attract score (656 legislator-term observations, 5.7%). Likewise, legislators receive an NA offer score if they did not offer any original cosponsorship in a term (31 legislator-term observations, 0.27%). Therefore, the resulting dataset includes an attract score for 10,893 legislator-term observations and an offer score for 11,518 legislator-term observations. Because the attract and offer scores are designed to measure bipartisanship through observed original cosponsorship decisions rather than legislators' broader orientation toward lawmaking, members who never attract or offer an original cosponsorship fall outside the scope of these measures; researchers interested in distinguishing such members can directly model legislative activity using the included `total_n_bills_sponsored` and `prop_bills_with_cosponsor` variables.

Calculating Issue-Specific Bipartisanship Measures

Given that users may be interested not only in aggregate bipartisanship, but also in whether legislators work across the aisle in specific issue areas (e.g., education, health, or transportation), we also construct attract and offer bipartisanship scores for each legislator-term across 34 policy issue areas. Table 2 reports the name of each policy issue area, the number of bills in that issue area, the percentage of all bills in that issue area, and an example bill. The CRS assigns each introduced bill a single policy topic based on the legislation's substantive content. This policy issue area captures the primary focus or predominant subject matter of each measure.⁵ To calculate within-issue-area attract and offer scores, we use the same formulas described above, but limit the cosponsorship and sponsorship decisions to bills within a given policy issue area.

⁵Our issue-specific scores capture the policy area of legislation at introduction and therefore may differ from the policy area through which a proposal's content ultimately becomes law. This, however, is consistent with our measurement goal: to determine whether legislators engage in bipartisanship within particular issue areas as original cosponsors at bill introduction. Thus, our issue-specific bipartisanship measures reflect legislators' intent to engage in bipartisanship within a particular issue area at introduction. As [40] and [41] show, provisions may later be incorporated into other legislative vehicles, including omnibus bills. Researchers studying downstream outcomes such as lawmaking success should therefore account for the possibility that legislative content may later advance through a different bill or policy area. The number of bills for which this is relevant, however, is relatively small: approximately 83% of bills in our sample never advanced beyond committee, increasing from 74.9% in the 98th Congress to 87.3% in the 117th and 118th Congresses. If bills do not advance beyond committee, their policy-area coding is unlikely to change, suggesting that even in an era of omnibus legislation, the overwhelming majority of legislation likely retains its original policy-area classification.

For example, health-specific attract scores measure out-party original cosponsorship on a legislator’s health bills, while health-specific offer scores measure a legislator’s original cosponsorship of out-party health bills. As with the aggregate scores, if a legislator did not attract or offer original cosponsorship within a particular policy issue area in a term, they receive an NA value. The dataset includes broader policy issue area coverage for bipartisanship offered because legislators are more likely to cosponsor other legislators’ bills than to sponsor bills themselves. The average legislator in the dataset has an offer score for more than half of the policy issue areas, but an attraction score for approximately 15% of policy issue areas.

Precision Weighting of Aggregate and Issue-Specific Bipartisanship Measures

One concern with our member-level bipartisanship measures is that they may be disproportionately influenced by very small numbers of bills or cosponsorship decisions. For example, a legislator who sponsors only one bill in a term and attracts bipartisan support on that bill would appear highly bipartisan under the raw measure, even though that score is based on only one bill. In contrast, a legislator who sponsors 100 bills and attracts bipartisan support on 40 is more bipartisan, despite receiving a lower raw proportion. The same concern applies to issue-specific scores, where the number of relevant bills or cosponsorships within a given policy area may be especially small. In short, raw bipartisanship scores are noisier when estimated from fewer observations and more precise when estimated from more observations.

To address this concern, we use a Bayesian shrinkage estimator to compute and apply weights to both the aggregate and issue-specific bipartisanship measures. This approach balances estimating parameters for groups (legislators) against a global population mean (all legislator scores in a given chamber-term) [42]. The intuition is straightforward: if a legislator’s score is based on only a few bills or cosponsorships, that score is less reliable and should be adjusted more heavily toward the average bipartisanship score for legislators in the same congressional term and chamber. If the score is based on many observations, it is more precise and should receive little to no adjustment.

For the aggregate attraction and offer scores, we define the prior target, μ , as the mean bipartisanship level within the same term and chamber. We then compute a precision weight, $w = n/(n + k)$, where n is the number of bills sponsored (cosponsored) by the legislator in that term and chamber and k is a fixed hyperparameter that determines how strongly the score is pulled toward the prior mean. The shrunk estimate $\hat{\theta}$ is:

$$\hat{\theta} = w \cdot \theta_{\text{obs}} + (1 - w) \cdot \mu \quad (3)$$

where θ_{obs} is the legislator’s observed score and μ is the Congress $\times$ chamber prior mean. We set $k = 10$, treating the prior as equivalent to 10 bills’ worth of information. This decision reflects a substantive judgment that a legislator requires approximately 10 sponsored bills before their observed score is considered more informative than the chamber-term mean. This choice is consistent with the broader empirical literature; the hyperparameter is fixed at a value that reflects the minimum number of observations needed to trust a raw proportion over a prior [43, 44]. In the U.S. Congress, 10 bills represent a modest threshold: most legislators sponsor well in excess of this number in a given term, meaning the prior exerts meaningful influence only for the least active members, which aligns with our primary measurement concern. Moreover, our results are not driven by our assigned hyperparameter value ($k = 10$); measures are consistent when we recalculate weighted scores with the hyperparameter set to $k = 5$ and $k = 15$. As a result, legislators with many sponsored or cosponsored bills receive weights close to 1 and retain scores close to their observed values. In contrast, legislators with fewer bills receive smaller weights and are pulled more strongly toward the Congress- and chamber-specific baseline.

We extend this same procedure to the issue-specific bipartisanship measures. In this case, the prior target, (μ), is defined more narrowly as the mean bipartisanship score within the same Congress, chamber, and policy topic. For topic-specific attraction scores, the precision term (n) is the number of bills the legislator sponsored within that topic; for topic-specific offer scores, the precision term (n) is the number of bills the legislator cosponsored within that topic. Because the number of observations is naturally smaller within individual policy areas, we set $k = 2$, treating the issue-specific prior as equivalent to two bills’ worth of information. Empirically, this threshold reflects the sparsity of within-issue legislative activity: two observations fall at approximately the 80th percentile of the issue-specific sponsorship distribution and the 53rd percentile of the issue-specific original-cosponsorship distribution. Thus, the issue-specific prior is most consequential for low-activity legislator-policy-area observations, where raw proportions are least reliable, while scores based on larger numbers of sponsored or cosponsored bills are weighted closer to a value of one, which retains their raw observed value.

Table 2. Policy Issue Area Coverage, Frequency, and Examples

Policy Issue Area	# of Bills	% of Total Bills	Example Bill (Congress)
Agriculture and Food	4,430	3.00%	H. R. 6413 (98th): A bill to direct the Secretary of Agriculture to carry out an acreage limitation program and a land diversion program for the 1985 crop of feed grains.
Animals	826	0.56%	S. 1368 (99th): A bill to end shipment of padded-jaw or steel-jaw leghold traps.
Armed Forces and National Security	10,926	7.40%	H. R. 4654 (100th): A bill to authorize the establishment of a National Guard in and for American Samoa.
Arts, Culture, Religion	429	0.29%	S. 2636 (101st): A bill to authorize the Board of Regents of the Smithsonian Institution to plan and design an extension of the National Air and Space Museum at Washington Dulles International Airport, and for other purposes.
Civil Rights and Liberties, Minority Issues	1,024	0.69%	H. R. 3583 (102nd): To extend the statute of limitation applicable to the filing of administrative complaints by Federal employees who allege employment discrimination in violation of title VII of the Civil Rights Act of 1964.
Commemorations	1,670	1.13%	S. 2200 (100th): A bill to amend Public Law 90-498 to provide for the designation of National Hispanic Heritage Month.
Commerce	4,328	2.93%	H. R. 4173 (103rd): To amend the Small Business Act to provide for expanded participation in the microloan demonstration program.
Congress	1,448	0.98%	S. 1039 (104th): A bill to require Congress to specify the source of authority under the United States Constitution for the enactment of laws, and for other purposes.
Crime and Law Enforcement	8,074	5.47%	H. R. 4443 (105th): To provide for the automatic revocation of the license of any licensed firearms dealer who willfully sells a firearm to a minor.
Economics and Public Finance	2,240	1.52%	S. 1097 (106th): A bill to offset the spending contained in the fiscal year 1999 emergency supplemental appropriations bill in order to protect the surpluses of the social security trust funds.
Education	6,225	4.21%	H. R. 2982 (106th): To provide grants to States and local educational agencies to recruit, train, and hire 100,000 school-based resource staff to help students deal with personal state of mind problems.
Emergency Management	1,657	1.12%	S. 1631 (107th): A bill to amend the Robert T. Stafford Disaster Relief and Emergency Assistance Act to direct the Director of the Federal Emergency Management Agency to conduct a study to determine the resources that are needed for development of an effective nationwide communications system for emergency response personnel.
Energy	4,046	2.74%	H. R. 1874 (107th): To allow any business or individual in any State experiencing a power emergency to operate any type of power generation available to ensure their economic stability, and for other purposes.
Environmental Protection	4,621	3.13%	S. 1716 (108th): A bill to amend the Federal Water Pollution Control Act to authorize the use of funds made available for nonpoint source management programs for projects and activities relating to the development and implementation of phase II of the storm water program of the Environmental Protection Agency.
Families	1,101	0.75%	H. R. 4347 (108th): International Assistance to Missing and Exploited Children Act of 2004.
Finance and Financial Sector	5,011	3.39%	S. 4116 (109th): A bill to amend the Federal Deposit Insurance Act, to clarify the scope of provisions relating to applicable rates of interest and other charge limitations.
Foreign Trade and International Finance	4,116	2.79%	H. R. 466 (109th): To amend the Trade Act of 1974 to delegate to the Under Secretary of Commerce for International Trade the functions relating to trade adjustment assistance for firms, and for other purposes.
Government Operations and Politics	10,610	7.18%	S. 3665 (110th): A bill to amend chapter 63 of title 5, United States Code, to modify the rate of accrual of annual leave for administrative law judges, contract appeals board members, and immigration judges.
Health	16,767	11.35%	H. R. 7139 (110th): To amend titles XVIII and XIX of the Social Security Act with respect to the qualification of the director of food services of a Medicare skilled nursing facility or a Medicaid nursing facility.
Housing and Community Development	2,035	1.38%	S. 3700 (111th): A bill to increase the maximum mortgage amount limitations under the Federal Housing Administration mortgage insurance programs for multi-family housing projects with elevators and for extremely high-cost areas.
Immigration	3,683	2.49%	H. R. 6080 (111th): Making emergency supplemental appropriations for border security for the fiscal year ending September 30, 2010, and for other purposes.
International Affairs	6,096	4.13%	S. 3366 (112th): A bill to designate the Haqqani network as a foreign terrorist organization.
Labor and Employment	4,396	2.98%	H. R. 5981 (112th): To amend title IV of the Employee Retirement Income Security Act of 1974 to provide for a guarantee by the Pension Benefit Guaranty Corporation for qualified preretirement survivor annuities under insolvent or terminated multiemployer pension plans.
Law	1,997	1.35%	S. 1207 (113th): Cameras in the Courtroom Act.
Native Americans	1,993	1.35%	H. R. 4694 (113th): To amend the Claims Resolution Act of 2010 to authorize the Secretary of the Interior to contract with eligible Indian tribes to manage land buy-back programs, to require that certain amounts be deposited into interest bearing accounts, and for other purposes.
Private Legislation	288	0.19%	S. 2474 (113th): A bill for the relief of Meriam Yahya Ibrahim, Martin Wani, and Maya Wani.
Public Lands and Natural Resources	7,950	5.38%	H. R. 5771 (114th): Conserving Our Reefs and Livelihoods Act of 2016.
Science, Technology, Communications	3,168	2.14%	S. 572 (115th): A bill to require the Secretary of Commerce to study the coverage gaps of the Next Generation Weather Radar of the National Weather Service and to develop a plan for improving radar coverage and hazardous weather detection and forecasting, and for other purposes.
Social Sciences and History	68	0.05%	H. R. 1088 (115th): African Burial Ground International Memorial Museum and Educational Center Act.
Social Welfare	3,203	2.17%	S. 4091 (116th): Emergency Aid for Returning Americans Affected by Coronavirus Act.
Sports and Recreation	297	0.20%	H. R. 6931 (116th): Professional Boxing Safety Enhancement Act of 2020.
Taxation	15,082	10.21%	S. 5262 (117th): A bill to amend the Internal Revenue Code of 1986 to extend the time during which a qualified disaster may have occurred for purposes of the special rules for personal casualty losses.
Transportation and Public Works	6,048	4.10%	H. R. 9648 (117th): Light Rail Transit Act.
Water Resources Development	1,794	1.21%	S. 1233 (118th): A bill to amend the Consolidated Farm and Rural Development Act to modify provisions relating to rural decentralized water systems grants.
NA	22	0.01%	H. R. 9547 (117th): To designate the facility of the United States Postal Service located at 120 Doty Street in Kaukauna, Wisconsin, as the "Sgt. Nickolas Mueller Post Office Building".
Total	147,669	100%	

We include both raw and precision-weighted attract and offer scores for all legislator-term observations. The correlation between the raw and weighted scores is quite high (Attract Correlation Coefficient = 0.90, Offer Correlation Coefficient = 0.99). This suggests that most legislators in the dataset sponsor and cosponsor close to the average number of bills. Therefore, the precision weighting addresses an important but relatively limited source of measurement error. We include both versions in the dataset because users may prefer one over the other depending on their research goals. Users may prefer the unweighted scores when they want the closest possible measure of observed bipartisan behavior in a given legislator-term, without any adjustment toward a chamber-specific prior mean. This may be especially useful when researchers want to apply their own weighting scheme, impose a different prior information threshold, or directly model the consequences of sparse legislative activity. In contrast, users may prefer weighted scores when they want measures that are less sensitive to instability arising from very small numbers of sponsored or cosponsored bills, especially when using our issue-specific scores, which are often based on fewer sponsor and cosponsorship decisions.

Data Record

The data are accessible through two platforms: Harvard Dataverse [37] and the R package `biparty` [38]. Both platforms provide the aggregate and policy issue-area bipartisanship score datasets. Harvard Dataverse additionally contains the underlying sponsorship and cosponsorship dataset, the legislator identifier crosswalk, and the crosswalk linking CRS policy areas to Comparative Agendas Project topic codes. We provide the aggregate and policy issue-area score datasets separately because users may prefer to work with them independently, though they can be merged using shared legislator identifiers. The data are available as comma-separated values (`.csv`) files and in R Data Serialization (`.rds`) format to facilitate broad reuse across statistical software, including Stata, R, and Python.

Aggregate Bipartisanship Scores: Each observation in this dataset is a legislator-term. Legislators appear multiple times if they served across multiple Congresses. The main variables of interest are `attract_aggregate_(weighted OR nonweighted)` and `offer_aggregate_(weighted OR nonweighted)`. The dataset includes both raw and precision-weighted versions of each bipartisanship measure, as well as legislator names, chamber, Congress, and party identification. Because users may also be interested in legislators’ overall sponsorship activity without wanting to visit the raw sponsorship and cosponsorship merged dataset, we include two additional variables: one measuring the total number of bills each legislator sponsored in a term and another capturing the proportion of those bills that attracted at least one cosponsor (regardless of party). We also include legislator-specific crosswalks with multiple unique identifiers (e.g., `govtrack_id`) to facilitate merging with other congressional data sources. Table 3 displays variable names, descriptions, and a sample observation from the aggregate bipartisanship scores dataset.

Table 3. Variables from Aggregate Dataset

Variable	Description	Example
<code>congress</code>	Congressional term in which the legislator served	117
<code>chamber</code>	Chamber in which the legislator served	SENATE
<code>name</code>	Legislator’s name	SUSAN COLLINS
<code>party</code>	Legislator’s party identification	R
<code>state</code>	State represented by the legislator	ME
<code>district</code>	Congressional district represented by the legislator (voting lower house members only)	No Congressional District (Senate)
<code>bioguide_id</code>	Unique legislator identifier from the Biographical Directory of the U.S. Congress	C001035
<code>govtrack_id</code>	Unique legislator identifier from GovTrack	300025
<code>thomas_id</code>	Legacy legislator identifier from the Library of Congress THOMAS system	Unavailable
<code>icpsr_id</code>	Standardized legislator identifier from ICPSR	49703
<code>wikipedia</code>	Unique identifier for the legislator’s Wikipedia page	Susan Collins
<code>wikidata</code>	Unique identifier for the legislator in Wikidata	Q22279
<code>attract_aggregate_weighted</code>	Precision-weighted aggregate bipartisanship attract score	0.758
<code>offer_aggregate_weighted</code>	Precision-weighted aggregate bipartisanship offer score	0.752
<code>attract_aggregate_unweighted</code>	Raw aggregate bipartisanship attract score	0.843
<code>offer_aggregate_unweighted</code>	Raw aggregate bipartisanship offer score	0.766
<code>total_n_bills_sponsored</code>	Total number of bills introduced	43
<code>prop_bills_with_cosponsor</code>	Proportion of bills introduced with at least one cosponsor (regardless of party)	0.977

Policy Issue Area Bipartisanship Scores: Each row is a legislator-term. Legislators therefore appear multiple times if they served in more than one congressional term. The main variables are issue-area scores measuring the extent to which legislators attract and offer original out-party cosponsorships within each of the 34 policy issue areas. Issue-area scores are stored in separate issue-specific columns. The dataset includes raw and precision-weighted versions of each issue-specific measure. In total, the dataset includes 140 score variables: 35 policy-area categories (34 substantive issue areas and one NA category) ×

two measures (attract and offer) $\times$ two versions (weighted and unweighted). Like the aggregate bipartisanship scores dataset, this dataset also includes unique legislator identifiers, chamber, Congress, and party identification. Table 4 displays variable names, descriptions, and a sample observation from the policy issue area bipartisanship scores dataset.

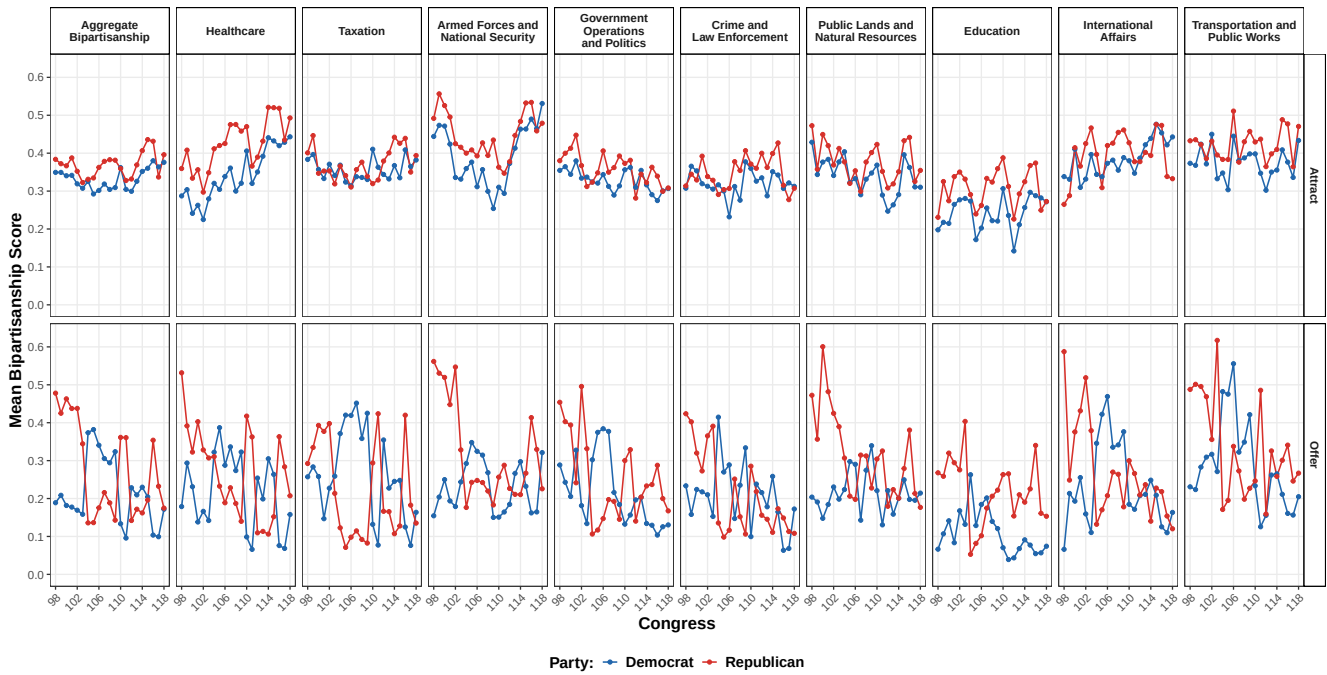
Table 4. Variables from Policy Issue Area Dataset

Variable	Description	Example
congress	Congressional term in which the legislator served	117
chamber	Chamber in which the legislator served	HOUSE
name	Legislator's name	ELISSA SLOTKIN
party	Legislator's party identification	D
state	State represented by the legislator	MI
district	Congressional district represented by the legislator	8
bioguide_id	Unique legislator identifier from the Biographical Directory of the U.S. Congress	S001208
govtrack_id	Unique legislator identifier from GovTrack	412784
thomas_id	Legacy legislator identifier from the Library of Congress THOMAS system	11082
icpsr_id	Standardized legislator identifier from ICPSR	21965
wikipedia	Unique identifier for the legislator's Wikipedia page	Elissa Slotkin
wikidata	Unique identifier for the legislator in Wikidata	Q30323721
attract_issuearea_weighted (n = 35)	Precision-weighted issue area bipartisanship attract score	0.399
offer_issuearea_weighted (n = 35)	Precision-weighted issue area bipartisanship offer score	0.0342
attract_issuearea_unweighted (n = 35)	Raw issue area bipartisanship attract score	0.6
offer_issuearea_unweighted (n = 35)	Raw issue area bipartisanship offer score	0

Data Overview

To illustrate how these scores can be used, Figure 2 plots average bipartisanship scores for both attract (top) and offer (bottom) measures, by party (line color and type), across the full time series (1983 – 2024). Across all panels, the y-axis represents the mean bipartisanship score and the x-axis denotes the congressional term. The figure includes aggregate scores and scores for nine selected policy issue areas, demonstrating how users can compare patterns of bipartisanship across time, parties, measures, and policy domains. Figure 2 highlights several features of the dataset that may be useful for future research. First, because the attract and offer measures capture distinct forms of cross-party collaboration, users can examine whether legislators' ability to attract out-party support follows patterns similar to their willingness to offer it. For example, Figure 2 demonstrates that aggregate bipartisanship attraction is relatively stable across the time series, while bipartisanship offered is more volatile. Second, users can compare aggregate scores with issue-specific scores to evaluate whether cross-party collaboration varies across policy domains. Figure 2 illustrates substantial variation in bipartisanship across policy domains. Among the issue areas displayed, Transportation and Public Works exhibits particularly high average bipartisanship across both attracting and offering out-party support (Attract = 11/34; Offer = 8/34), whereas Education exhibits comparatively low levels across both dimensions (Attract = 30/34; Offer = 32/34). Third, users can summarize trends by party across congressional terms, allowing them to assess whether changes in bipartisan activity are concentrated in particular parties, time periods, or issue areas. The purpose of Figure 2 is illustrative rather than inferential. It provides a descriptive overview of the data's structure and scope and shows how researchers can use the aggregate and issue-specific scores to generate comparisons across multiple dimensions of congressional bipartisanship. Future users may extend this approach by examining additional policy areas, comparing chambers, studying individual legislators, merging the scores with external covariates, or using the data to study the causes and consequences of bipartisan collaboration in Congress.

Figure 2. Attract and Offer Bipartisanship Scores by Party and Issue Area



Technical Validation

Internal Validation

To validate the accuracy of the data scraped from `congress.gov`'s API, we employed a team of research assistants to hand-code 5,000 randomly sampled cosponsor-bill pairs. We expected to find few, if any, errors because the data were scraped from `congress.gov` rather than manually labeled or entered. However, we viewed this validation step as necessary because incorrect information across these variables would affect our final scores.

Each row in the validation set includes the bill sponsor's name and the name of a single cosponsor, drawn from the full list of cosponsors for that bill. As a result, each research assistant validated one cosponsor-bill pairing rather than the complete set of cosponsors for a given bill. The validation protocol required research assistants to verify five variables: (1) total cosponsor volume, (2) bill sponsorship date, (3) cosponsorship date, (4) sponsor party, and (5) cosponsor party. These are the five variables integral to the creation of the scores. For each observation, research assistants independently navigated to the corresponding bill page on `congress.gov`, recorded the values they observed, and submitted their responses for evaluation against the scraped data. Research assistants were trained to code the validation set during a one-hour meeting. They then completed a training set of 50 observations, which the project PIs reviewed before providing feedback. Upon successful completion of the training set, research assistants received the full validation set.

Overall, as expected, the validation process shows that information scraped from the `congress.gov` API is highly consistent with information available locally on `congress.gov`. We report five validation statistics assessing alignment between the scraped API data and the local `congress.gov` records: Cohen's Kappa Coefficient, average error rate, precision, recall, and F1 score. For all five variables, Cohen's kappa is at or above 0.998, indicating a high level of agreement between the two data sources. Precision, recall, and F1 scores are similarly high. The highest average error rate is 0.002, indicating that very few observations were misaligned across the two sources. Collectively, these results suggest that the `congress.gov` API is a reliable and accurate source for collecting bill-level and cosponsor-level data from `congress.gov`.

One important exception, however, was an inconsistency between `congress.gov`'s original cosponsor designation—a logical indicator returned by the congressional API (e.g., `TRUE/FALSE`)—and the underlying sponsorship and cosponsorship dates returned by the same API. The original cosponsor designation from `congress.gov` should be coded as `TRUE` only when the cosponsorship date matches the sponsorship date [45, 46]. In several cases, however, the original cosponsorship

designation was coded as `TRUE` even when the cosponsorship and sponsorship dates did not match. This discrepancy is consequential for score creation because the original cosponsor designation provided by `congress.gov` is commonly used to identify original cosponsorship. For our purposes, accurately identifying original cosponsors is integral to score construction. To address this issue, we construct our own date-based indicator of original cosponsorship using the sponsorship and cosponsorship date fields, coded as 1 when the dates match and 0 otherwise.

The `congress.gov` original-cosponsor designation identifies 920,281 cosponsorship decisions as original, compared to 891,521 identified by our date-based measure. Across 2,486,089 total cosponsorship decisions, `congress.gov` incorrectly classifies 28,760 cosponsorships as original even though the underlying date fields indicate that they occurred after the bill's sponsorship date. Although modest in percentage terms (1.16%), this error is systematic: because our dataset spans decades and hundreds of thousands of bills, even small misclassification rates compound into meaningful measurement error at scale.

Table 5. Variables and Validation Statistics from Cosponsorship Dataset

Measure	Description	# Cosponsors	Sponsor Party	Sponsorship Date	Cosponsor Party	Cosponsorship Date
Cohen's Kappa Coefficient	Measures agreement between human coders and the <code>congress.gov</code> API after adjusting for chance agreement.	0.999	0.999	0.999	0.999	0.998
Average Error Rate	Measures the share of compared observations that were miscoded by human coders relative to the <code>congress.gov</code> API.	0.001	0.000	0.001	0.000	0.002
Precision	Measures the average one-vs-rest precision across observed values, indicating how often human-coded values matched the corresponding <code>congress.gov</code> API values (i.e., minimizing false positive).	0.992	1.000	0.998	1.000	0.997
Recall	Measures the average one-vs-rest recall across observed values, indicating how completely human coders recovered the corresponding <code>congress.gov</code> API values (i.e., minimizing false negative).	1.000	1.000	0.998	1.000	0.998
F1 Score	Measures the average one-vs-rest harmonic mean of precision and recall, summarizing agreement between human coders and the <code>congress.gov</code> API.	0.998	1.000	0.999	1.000	0.999

External Validation

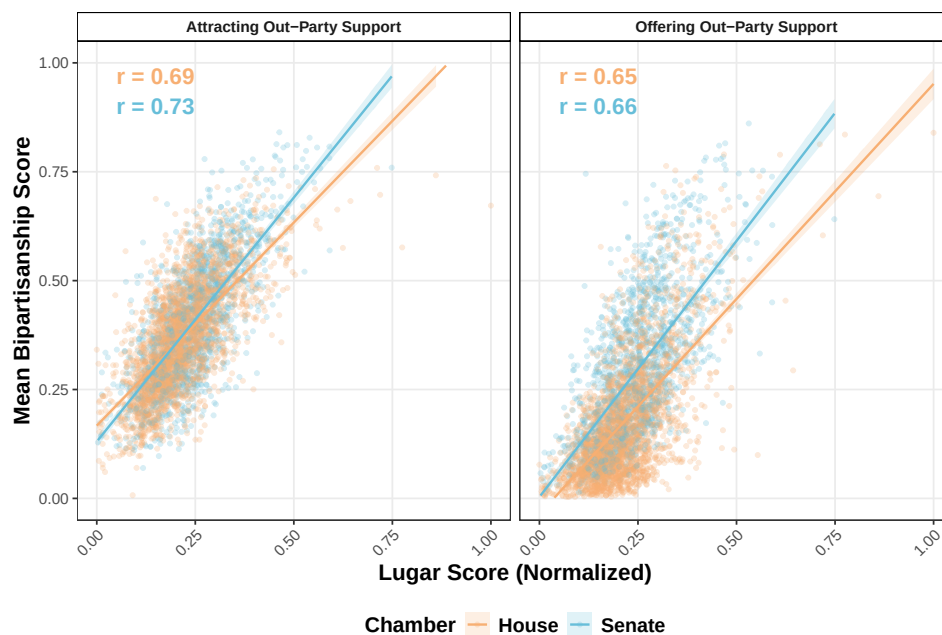
To assess the construct validity of our bipartisanship scores, we validate them against the Lugar Center's Bipartisan Index [36]. The Lugar Center's index is among the most widely cited measures of bipartisanship in Congress, making it a natural benchmark against which to evaluate our scores. Because the Lugar Center publishes its scores as PDFs rather than in a machine-readable format, we manually collected the scores. Lugar Center scores are available for the Senate beginning with the 103rd Congress (1993–1995) and for the House beginning with the 113th Congress (2013–2015). We normalize these scores before comparison to place both measures on a common scale (0–1). Importantly, the Lugar Center excludes members who served fewer than 10 months of a full Congress (and fewer than six months when scores are reported for a single calendar year), as well as chamber leaders.

The Lugar Center's Bipartisan Index differs from our scores in four important ways: it has more limited time-series coverage, does not distinguish between attracting and offering out-party support, is not sensitive to cosponsorship timing, and does not weight scores by a member's overall sponsorship and cosponsorship activity within a congressional term. In contrast, our scores cover the 98th through 118th Congresses and provide two distinct measures of bipartisanship—*attract* and *offer*—based only on original out-party cosponsorship. Because the Lugar Center's Bipartisan Index is not calculated by policy issue area, we compare only our aggregate measures to their index. We expect our measures to be correlated with the Lugar Center's scores because both capture bipartisan behavior. However, given the measurement differences described above, we do not expect the correlations to be perfect. Table 6 reports the term-level correlations between the Lugar Center scores and our aggregate *attract* and *offer* measures.

Table 6. Correlation Between Offer and Attract Bipartisanship Scores and Lugar Score by Congress

Congress	Lugar & Attracting Out-Party Support	Lugar & Offering Out-Party Support
103	0.475	0.566
104	0.657	0.723
105	0.677	0.721
106	0.455	0.675
107	0.539	0.351
108	0.727	0.591
109	0.679	0.549
110	0.808	0.657
111	0.746	0.721
112	0.828	0.773
113	0.687	0.686
114	0.666	0.730
115	0.729	0.780
116	0.705	0.522
117	0.762	0.647
118	0.768	0.818

The average correlation across terms is 0.682 for attraction and 0.657 for offer. The highest correlation occurs between the Lugar scores and our attraction measure in the 112th Congress (0.828), while the lowest correlation occurs between the Lugar scores and our offer measure in the 107th Congress (0.351). Figure 3 plots the correlations between the Lugar scores and our attraction and offer measures separately by chamber. Our measures are slightly more highly correlated with the Lugar Center’s Senate scores, though the cross-chamber differences are modest. Collectively, these results suggest that our measures capture the same latent concept as the Lugar Center’s index, bipartisanship, while also reflecting meaningful differences in measurement strategy.

Figure 3. Correlation Between Offer and Attract Bipartisanship Measures and the Lugar Center Bipartisanship Measure by Chamber

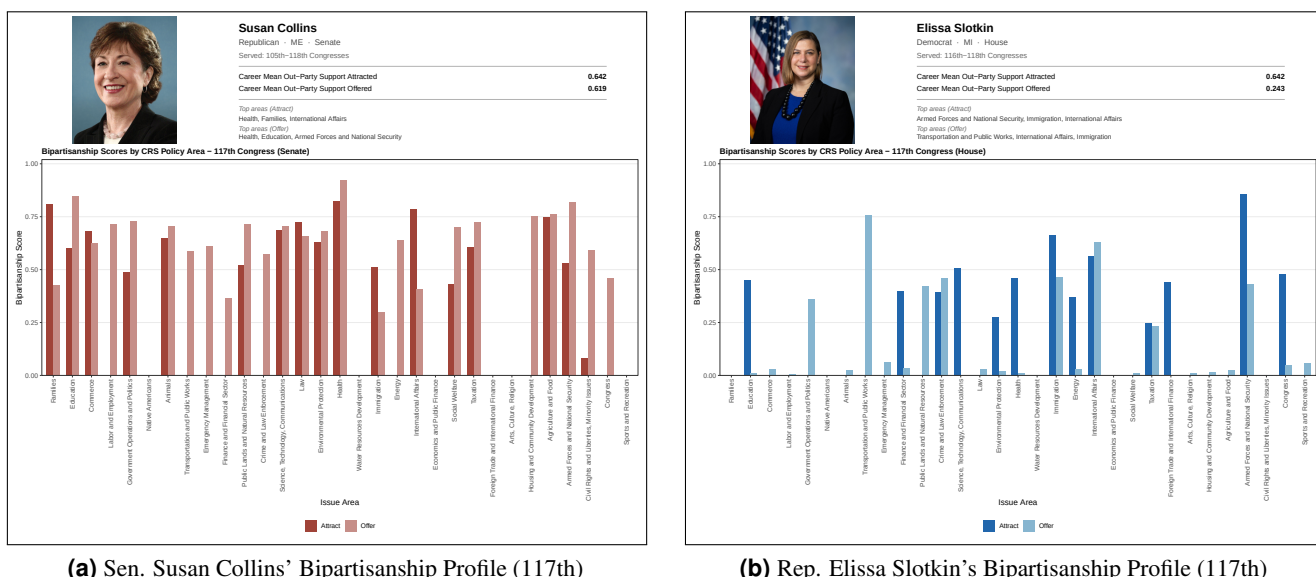
Usage Notes

The unit of analysis in both the aggregate and policy issue area datasets is the legislator-term. We do not merge these datasets because users may prefer to use them independently. However, because both datasets share the same unit of analysis, users can merge them if desired. To facilitate merging the two datasets, as well as linking our data to other congressional covariates, we include a crosswalk with several unique legislator identifiers: `bioguide`, `govtrack`, `THOMAS`, and `ICPSR`. We encourage users to rely primarily on `bioguide` and `govtrack` when possible. Other identifiers have important limitations. `ICPSR` excludes legislators who served only a partial term and non-voting delegates. The `THOMAS` identifier is available only for House members. We include all of these identifiers because few datasets aggregate them into a single crosswalk, and users may need to rely on a specific identifier depending on the structure of the data they are merging with ours.

To facilitate broad use among researchers, journalists, legislators, and the public, we host the data on both Harvard Dataverse [37] and an R package, `biparty` [38]. Scores will be updated in both locations at the end of each congressional term. The R package is designed for users who regularly work with large datasets. The package loads both datasets and includes functions that allow users to retrieve and compare member scores, analyze bipartisanship by legislator, issue area, state, term, and chamber, and merge the data with other relevant covariates. Individual member data can be accessed through several lookup functions: users can retrieve a single legislator's attract and offer scores (`get_member_scores()`), return scores across all Congresses in which they served (`get_member_trend()`), produce side-by-side comparisons for a user-specified set of members (`compare_members()`), or retrieve scores for all members from a given state (`get_state_delegation()`). Aggregate summaries are available at the congressional level via `get_congress_summary()`, with optional breakdowns by party, chamber, and issue area.

Issue-area analyses are supported by functions that return all legislators' scores for a given policy area in a specified Congress (`get_issue_scores()`), track how mean bipartisanship across the attract and offer dimensions has changed over time (`get_issue_trend()`), and enumerate all 34 CRS policy area labels with their corresponding full-length names as a crosswalk for use across package functions (`list_issues()`). The package also includes a suite of visualization tools: users can generate a violin plot of score distributions across issue areas (`plot_bipartisanship()`), render a time-series line chart for one or more members or parties (`plot_trend()`), and generate a member profile card combining a legislator photograph with career-level scores, top issue areas, and an issue-area bar chart for a specified Congress (`plot_member_profile()`). Figure 4 provides two example profile cards from the 117th Congress (2021 - 2023): Senator Susan Collins (R - ME) and Representative Elissa Slotkin (D - MI). More information about using the `biparty` package can be found [here](#).

Figure 4. Example Member Profile Cards from `biparty`



Note: Member photographs are official public-domain congressional photographs retrieved through the `unitedstates/images`

repository, which sources the images from the U.S. Government Publishing Office.

Data Availability

All data produced in this study are publicly available through [Harvard Dataverse](#) [37]. The repository includes (1) the aggregate bipartisanship scores dataset; (2) the policy issue area bipartisanship scores dataset; (3) the merged sponsorship and cosponsorship data from which the scores are constructed; (4) the underlying bill, sponsor, cosponsor, and policy area datasets; (5) the legislator identifier crosswalk; and (6) the crosswalk linking CRS policy areas to Comparative Agendas Project topic codes. All datasets are provided in comma-separated values (.csv) and R Data Serialization (.rds) formats and are released under a Creative Commons Attribution 4.0 International (CC BY 4.0) license. The Comparative Agendas Project topic codes included in the policy area crosswalk are drawn from the Policy Agendas Project's Congressional Bills dataset and should be cited as requested by its creators [39]. The score datasets are also available through the R package [biparty](#) [38].

Code Availability

The replication code used to clean, merge, and construct the bipartisanship scores is publicly available through [Harvard Dataverse](#) [37]. The R package [biparty](#), which provides functions for retrieving, analyzing, and visualizing the scores, is available through GitHub [38]. The scraping, cleaning, and data transformation were performed using R (v4.5.1) [47] on a Mac Studio with an Apple M4 Max chip (14-core CPU: 10 performance and 4 efficiency cores) and 36 GB of memory, running macOS Tahoe 26.2 (aarch64-apple-darwin20). Issue area labels were scraped using Python (v3.12.11) [48] in the Spyder IDE (v6.1.2) [49].

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Author Contributions Statement

MRD and JML scraped the data, created the measures, validated the scores, developed the R package, and wrote the manuscript.

Competing Interest

The authors declare no competing interests.

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